

Spatio-Temporally Regularized Common Spatial Patterns (STR-CSP) for Single-Trial EEG Classification

Feifei Qi¹, Yuanqing Li¹, Wei Wu¹ *

1. School of Automation Science and Engineering, South China University of Technology, Guangzhou 510640, China.

Asterisk indicates corresponding author.

E-mail: auweiwu@scut.edu.cn

Abstract: Common Spatial Patterns (CSP) is among the most popular spatial filtering methods used in motor imagery based brain-computer interfaces (BCIs). In this paper, a novel algorithm termed *spatio-temporally regularized common spatial patterns* (STR-CSP) is proposed to enhance CSP such that discriminative spatio-temporal EEG patterns can be extracted. Unlike most previous CSP-based spatio-temporal filtering algorithms in which spatial and temporal filters are optimized in a sequential manner, STR-CSP optimizes spatio-temporal filters simultaneously under a unified criterion with analytical solutions. Moreover, the algorithm enables us to alleviate the over-fitting problem by l_2 -norm regularization. Successful application to the analysis of a BCI competition EEG data set comprising 18 subsets demonstrates the superiority of STR-CSP over contemporary methods.

Key Words: Brain-Computer Interface (BCI), electroencephalography (EEG), motor imagery (MI), common spatial patterns (CSP), combined spatio-temporal filter, l_2 -norm regularization

1 Introduction

Brain-computer interfaces (BCIs), which aim at translating brain signals into control commands for output devices [1], have been under intensive studies in recent years. In EEG-based BCI systems using the motor imagery paradigm [2], discriminative information is captured via quantifying band-power decrease, termed *event related de-synchronization* (ERD), within mu (7-15 Hz) and beta (15-30 Hz) rhythms. However, scalp EEG signals are severely distorted by various sources of noise and suffer from low spatial resolution due to volume conduction, giving only a blurred representation of cortical or subcortical brain activities. *Common spatial patterns* (CSP) [3]–[5] is a popular and efficient spatial filtering algorithm to improve the signal-to-noise ratio (SNR) of EEG signals. Given EEG signals recorded under two states of motor imageries, CSP can be employed to extract spatial patterns most discriminative between the states. Let C denote the number of channels, then CSP can be cast as a Rayleigh quotient maximization problem [6]:

$$\max_{\mathbf{w}} J_1(\mathbf{w}) = \frac{\mathbf{w}^\top \mathbf{R}_1 \mathbf{w}}{\mathbf{w}^\top \mathbf{R}_2 \mathbf{w}} \quad (1)$$

where $\mathbf{R}_1, \mathbf{R}_2 \in \mathbb{R}^{C \times C}$ are the average covariance matrices of the two states, and $\mathbf{w} \in \mathbb{R}^C$ is the spatial filter to be optimized. Equivalently, the spatial filter can be obtained by performing the generalized eigenvalue decomposition of $\mathbf{R}_1$ and $\mathbf{R}_2$ [7]: $\mathbf{R}_1 \mathbf{w} = \lambda \mathbf{R}_2 \mathbf{w}$. The resulting eigenvalue λ is often assumed to be a nice measure of the separability between the spatially filtered signals of the two states.

The use of CSP for EEG analysis requires the temporal filter pre-determined. Nonetheless, the range of the discriminative frequency band in EEG signals may vary across individuals. Ordinarily, the temporal filter is manually set to

encompass a broad passband based on generic prior knowledge (e.g., 8-30 Hz for ERD during motor imagery), which is imprecise. Alternatively, it can be determined via cross-validation, which is time-consuming. From a practical viewpoint, learning subject-specific and easy-to-implement temporal filters is critical for further boosting BCI's performance. Last decades witnessed the emergence of an array of CSP-based methods, focusing on designing filters to extract discriminative patterns in both spatial and temporal/frequency domains [9]–[13]. Some of them are briefly introduced below.

The *common spatio-spectral pattern* (CSSP) [9] algorithm embedded a first-order finite impulse response (FIR) temporal filter with the spatial filter, and both were optimized simultaneously. To increase the flexibility of temporal filtering, FIR filters of higher orders was proposed to be used in *common sparse spectral spatial pattern* (CSSSP) [10], in which regularization was incorporated to sparsify the solution so as to ameliorate overfitting. However, since analytical solutions were no longer available, CSSSP could only hope to obtain local optima by solving a non-convex optimization problem.

Several algorithms were proposed to optimize the spatial and temporal filters in separate stages. *Filter bank common spatial pattern* (FBCSP) [13] and its variants first decomposed the EEG data into an array of sub-band components with CSP applied to each component, and then a reduced set of features from those generated from all the components was selected for classification. *Spectrally weighted common spatial pattern algorithm* (SPEC-CSP) [12] and *iterative spatio-spectral patterns learning* (ISSPL) [11] updated the spatial and spectral filters alternatively, with the maximal fisher ratio and maximal margin hyper-plane employed as the criteria for optimizing spectral filters, while the spatial filters were obtained by CSP. However, the convergence of SPEC-CSP and ISSPL was not guaranteed.

This paper presents a novel algorithm termed *spatio-temporal regularized common spatial patterns* (STR-CSP), in which the spatial filters and temporal filters were simultaneously optimized under a single criterion, with analytical

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solutions available. Moreover, an l_2 -norm regularizer was incorporated in the objective function to address the overfitting issue associated with the high dimensionality of the parameter space. To verify its effectiveness, an experimental study was conducted to compare STR-CSP with three existing algorithms, namely CSP, FBCSP, and CSSP.

The reminder of the paper is organized as follows. The STR-CSP algorithm is derived in section II. Experimental results of the compared algorithms are presented in section III. Finally, section IV concludes the paper.

2 The STR-CSP algorithm

2.1 Spatio-temporal filter optimization

Learning high-order FIR temporal filters offers more flexibility and potentially better discriminability. The goal of STR-CSP is to optimize arbitrary-order FIR temporal filters in conjunction with spatial filters. In STR-CSP, a separate temporal filter is optimized for each channel, which is motivated by our working hypothesis that the discriminative frequency bands in the EEG signal are channel-dependent. In the following we show how STR-CSP optimizes combined spatio-temporal filters, by maximizing the ratio of transformed data's variances under the two states.

In analogy to CSSP [9], in which merely one time-delayed data copy was considered, we define our objective function based on multiple time-delayed data copies. Specifically, let N, T be the order of the temporal filter and the number of sample points, respectively, and $\mathbf{X} \in \mathbb{R}^{C \times T}$ be the EEG recording of a single trial. We first consider the case of a single spatial filter $\mathbf{w} = [w_1, w_2, \dots, w_C]^T \in \mathbb{R}^C$ along with its corresponding channel-wise temporal filters $\mathbf{a}_c = [a_{c1}, a_{c2}, \dots, a_{cN}]^T \in \mathbb{R}^N, c = 1, 2, \dots, C$. Formally, spatial filtering can be mathematically expressed via a *spatial filtering operator* $f_{\mathbf{w}}$:

$$\begin{aligned} f_{\mathbf{w}} : \mathbb{R}^{C \times T} &\longrightarrow \mathbb{R}^{1 \times T} \\ \mathbf{X} &\longrightarrow f_{\mathbf{w}}(\mathbf{X}) = \sum_{c=1}^C w_c \mathbf{X}_{c,\cdot} \end{aligned} \quad (2)$$

Similarly, for the c -th channel, temporal filtering can be expressed via the following *temporal filtering operator* $f_{\mathbf{a}_c}$:

$$\begin{aligned} f_{\mathbf{a}_c} : \mathbb{R}^T &\longrightarrow \mathbb{R}^T \\ \mathbf{X}_{c,\cdot} &\longrightarrow f_{\mathbf{a}_c}(\mathbf{X}_{c,\cdot}) = \sum_{n=1}^N a_{cn} \mathbf{X}_{c,\cdot}^{(n-1)} \end{aligned} \quad (3)$$

where $\mathbf{X}_{c,\cdot}^{(n-1)}$ denotes the data copy delayed by $n-1$ time steps for the c -th channel. For convenience, the temporal operators for all channels are compactly denoted by $f_{\mathbf{a}}$:

$$\bar{\mathbf{X}} = f_{\mathbf{a}}(\mathbf{X}) = \begin{pmatrix} f_{\mathbf{a}_1}(\mathbf{X}_{1,\cdot}) \\ f_{\mathbf{a}_2}(\mathbf{X}_{2,\cdot}) \\ \vdots \\ f_{\mathbf{a}_C}(\mathbf{X}_{C,\cdot}) \end{pmatrix} \quad (4)$$

Therefore, the combined operations of the spatial and temporal filtering can be collectively expressed via $f_{\mathbf{aw}}$:

$$\begin{aligned} f_{\mathbf{aw}} : \mathbb{R}^{C \times T} &\longrightarrow \mathbb{R}^{1 \times T} \\ \mathbf{X} &\longrightarrow f_{\mathbf{aw}}(\mathbf{X}) = f_{\mathbf{w}} \cdot f_{\mathbf{a}}(\mathbf{X}) = f_{\mathbf{w}}(\bar{\mathbf{X}}) \\ &= \sum_{c=1}^C w_c \sum_{n=1}^N a_{cn} \mathbf{X}_{c,\cdot}^{(n-1)} = \mathbf{s}_{\mathbf{aw}} \tilde{\mathbf{X}} \end{aligned} \quad (5)$$

where $\mathbf{s}_{\mathbf{aw}} = [w_1 a_{11}, w_2 a_{12}, \dots, w_C a_{1C}, \dots, w_1 a_{N1}, w_2 a_{N2}, \dots, w_C a_{NC}]^T$ denotes the spatio-temporal filter to be optimized, and

$$\tilde{\mathbf{X}} = \begin{pmatrix} \mathbf{X} \\ \mathbf{X}^{(1)} \\ \vdots \\ \mathbf{X}^{(N-1)} \end{pmatrix} \quad (6)$$

is an appended data matrix that adds the delayed signals as “new channels”, where $\mathbf{X}^{(n)}, n = 1, 2, \dots, N-1$ denote the original data delayed by n time steps.

According to (5), the spatio-temporal filtering operations can be represented by a single linear transformation of $\tilde{\mathbf{X}}$, with the transformation vector $\mathbf{s}_{\mathbf{aw}}$ encompassing both the spatial and temporal filters. In other words, $\mathbf{s}_{\mathbf{aw}}$ is a reparametrization of the spatio-temporal filters. Hence, a natural spatio-temporal filter optimization scheme is to first optimize $\mathbf{s}_{\mathbf{aw}}$ from $\tilde{\mathbf{X}}$ using CSP, and then decompose $\mathbf{s}_{\mathbf{aw}}$ into $\mathbf{w}$ and $\mathbf{a}_c$. More specifically, let $\tilde{\mathbf{R}}_1$ and $\tilde{\mathbf{R}}_2$ be the average covariance matrices for the two states calculated from $\tilde{\mathbf{X}}$. $\mathbf{s}_{\mathbf{aw}}$ can be obtained by solving the following optimization problem:

$$\max_{\mathbf{s}_{\mathbf{aw}}} J_2(\mathbf{s}_{\mathbf{aw}}) = \frac{\mathbf{s}_{\mathbf{aw}}^\top \tilde{\mathbf{R}}_1 \mathbf{s}_{\mathbf{aw}}}{\mathbf{s}_{\mathbf{aw}}^\top \tilde{\mathbf{R}}_2 \mathbf{s}_{\mathbf{aw}}} \quad (7)$$

Analogous to CSP, performing the generalized eigenvalue decomposition of $\tilde{\mathbf{R}}_1$ and $\tilde{\mathbf{R}}_2$ yields multiple ($\leq CN$) combined spatio-temporal filters for each state.

2.2 Regularization

The number of unknown parameters involved in (7) is CN , which is substantially increased compared with C in CSP. Thus the solution may be prone to overfitting the training data, which harms generalization. A convenient way of controlling the complexity of the combined spatio-temporal filters is to introduce regularization, by which the size of the parameter search space can be restricted properly.

In STR-CSP, the Tikhonov regularization, as also utilized in [14], is employed to seek a tradeoff between the capacity of the learning algorithm, as measured by the generalized Rayleigh quotient, and the complexity of the solution. With a weight matrix $\mathbf{K}$, the penalty function for the Tikhonov regularization is $P(\mathbf{s}_{\mathbf{aw}}) = \|\mathbf{s}_{\mathbf{aw}}\|_{\mathbf{K}}^2 = \mathbf{s}_{\mathbf{aw}}^\top \mathbf{K} \mathbf{s}_{\mathbf{aw}}$. In this paper, we let $\mathbf{K} = \mathbf{I}$, thus $P(\mathbf{s}_{\mathbf{aw}}) = \|\mathbf{s}_{\mathbf{aw}}\|_2^2$, which is the l_2 -norm penalty. What this implies is that the weight coefficients are penalized equally for the spatio-temporal filter. Apart from its simplicity, another advantage of using the l_2 -norm penalty is that the spatio-temporal filter can be optimized efficiently by essentially the same algorithm as the one without regularization.

By incorporating the l_2 -norm penalty in (7), STR-CSP strives to solve the following optimization problem:

$$\max_{\mathbf{s}_{\mathbf{aw}}} J_3(\mathbf{s}_{\mathbf{aw}}) = \frac{\mathbf{s}_{\mathbf{aw}}^\top \tilde{\mathbf{R}}_1 \mathbf{s}_{\mathbf{aw}}}{\mathbf{s}_{\mathbf{aw}}^\top (\tilde{\mathbf{R}}_2 + \rho \mathbf{I}) \mathbf{s}_{\mathbf{aw}}} \quad (8)$$

where ρ is the regularization parameter. The resulting combined spatio-temporal filters are the eigenvectors corresponding to the largest eigenvalues of $\mathbf{M}_1 = (\tilde{\mathbf{R}}_2 + \rho \mathbf{I})^{-1} \tilde{\mathbf{R}}_1$ for the first state, and the eigenvectors corresponding to the largest eigenvalues of $\mathbf{M}_2 = (\tilde{\mathbf{R}}_1 + \rho \mathbf{I})^{-1} \tilde{\mathbf{R}}_2$ for the second state.

3 Experimental results

3.1 The EEG data set for evaluation

We use one motor imagery EEG data set (Data Set IIIa, BCI competition III) to evaluate the proposed algorithm, and compare it with CSP, FBCSP and CSSP. The data set consists of EEG signals recorded from three subjects who performed left-hand, right-hand, foot, and tongue (labeled 1, 2, 3, and 4) motor imageries. The EEG signals were recorded using 60 channels, with a sampling rate of 250 Hz. Both training and testing sets contain 45 trials per class for subject $K3$, and 30 trials per class for subjects $K6$ and $L1$. To evaluate the single-trial binary classification performance, we split the four-category data of each subject to generate $C_4^2 = 6$ binary classification problems, thereby a total of $6 \times 3 = 18$ data subsets are obtained (denote by $K6-12$ the subset for which $K6$ performed left-hand and right-hand imageries, and so on).

3.2 Preprocessing

For fair comparisons of the compared algorithms, the identical preprocessing setting is applied to each data subset:

- channel selection (all 60 channels are used).
- band-pass filtering (7-30 Hz, using a sixth-order Butterworth filter). The passband restricts the analysis to the mu and beta rhythms, and eliminates the noise unrelated with sensorimotor rhythms.
- time segment (0.5-3.5 s after the cue instructing the subject to perform motor imageries).

Note that for FB-CSP, six sixth-order Butterworth filters were used to band-pass filter the EEG signals into six non-overlapping subband (7-11 Hz, 11-15 Hz, 15-19 Hz, 19-23 Hz, 23-27 Hz, and 27-30 Hz) components. $\mathbf{R}_1$, $\mathbf{R}_2$, $\tilde{\mathbf{R}}_1$, and $\tilde{\mathbf{R}}_2$ were estimated by averaging the covariance matrices of all EEG trials within a state.

3.3 Hyper-parameter selection

The hyper-parameters are the temporal filter's order (N) and regularization coefficient (ρ) for STR-CSP, and the delay time (τ) for CSSP. The candidates for the hyper-parameters are specified as follows: $\tau \in [0, 1, 2, 3, 4, 5, 6, 7, 8, 9]$ (when $\tau = 0$ CSSP is reduced to CSP), and $N \in [6, 8, 10]$, $\rho \in 10^{-6} \times [1, 2, \dots, 50]$.

3.4 Results and discussion

For each subset, the training EEG data were fed into the compared algorithms for learning spatial or spatio-temporal filters, with the hyper-parameters determined via 10-fold cross-validations. The log-variances of the filtered signals (3 for each state) were then computed as the input to train a Fisher linear discriminant analysis (FLDA) classifier. The classification accuracies obtained on the test sets are provided in Table 1, with the highest classification accuracies highlighted in boldface for each data subsets and the determined hyper-parameter values listed in the parentheses. The results indicate that STR-CSP performs the best for 15 out of the 18 subsets, with the mean classification accuracies surpassing CSP, CSSP, and FBCSP. Bonferroni-corrected Wilcoxon signed-rank tests show that the differences between the classification accuracies of the compared algorithms are significant at the 5% level.

In line with the results reported in previous studies, the

performance gain of STR-CSP over CSP again warrants the need of optimizing temporal filters to overcome the limitation of the broad-band temporal filter. In particular, STR-CSP outperforms CSP by 13% for $K6-24$. The 2% increase in the mean classification accuracy of STR-CSP over CSSP suggests that it is worth the effort to learn high-order FIR temporal filters as opposed to first-order filters. Moreover, it has been mentioned in [9] that incorporating more than one delay time will tend to overfit the data. Thus, to a large extent the performance enhancement should also be attributed to the introduction of the regularization in our problem formulation to prevent overfitting.

It can also be observed that the accuracy improvement of STR-CSP is the most conspicuous for some of the subjects poorly classified by CSP, CSSP, and FBCSP, e.g., subsets $K6-12$, $K6-24$, and $L1-34$. Subsets (e.g., subject $K3$) with already excellent accuracies by CSP, CSSP, and FBCSP exhibit no particular advantage of using STR-CSP since there was little room for improvement.

Fig.1 illustrates the weights of the spatial filters obtained by CSP, CSSP, and STR-CSP for subset $L1-12$. The weight distributions from STR-CSP over the scalp are physiologically more relevant to performed imagery tasks than those from CSP and CSSP, as demonstrated by the strong weights over the corresponding regions in the contralateral motor cortex. Furthermore, the log-variance features generated from STR-CSP are more correlated with the performed imagery task, as indicated by r^2 listed below each weight map.

4 Conclusions

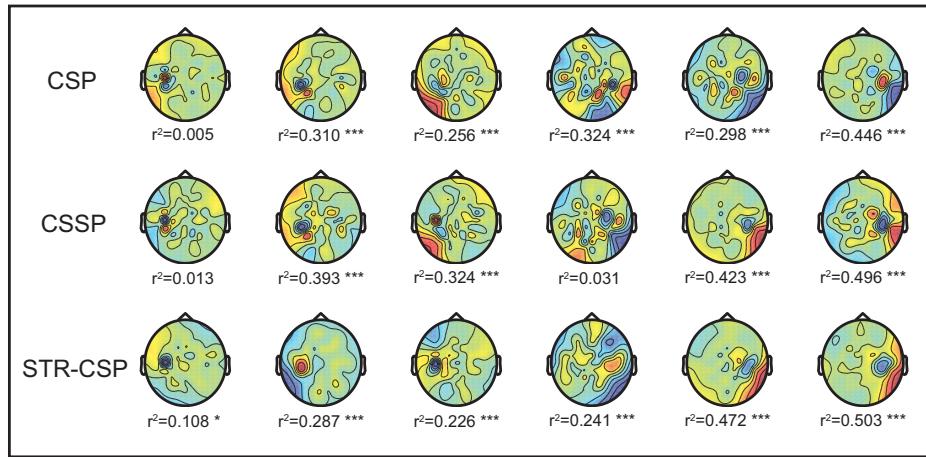
In this paper, we presented a novel spatio-temporal filter optimization algorithm termed STR-CSP, and applied it to binary classification of single-trial motor imagery EEG data. In the proposed algorithm, the spatial and temporal filters are jointly and efficiently optimized under a unified criterion. Moreover, the l_2 -norm penalty is incorporated in the objective function to guard against overfitting. Classification results on a BCI competition data set demonstrate the efficacy of STR-CSP and its superiority over CSP, FBCSP and CSSP, with more discriminative and neurophysiologically meaningful patterns extracted.

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Table 1: Comparisons of the classification accuracies (%) of CSP, FBCSP, CSSP and STR-CSP on Data Set IIIa from BCI competition III.

Subject	CSP	FBCSP	CSSP	STR-CSP
K3-12	97.78	92.22	98.89($\tau = 1$)	100.00 ($N = 10, \rho = 1 \times 10^{-6}$)
K3-13	98.89	94.44	98.89 ($\tau = 4$)	98.89 ($N = 10, \rho = 1 \times 10^{-6}$)
K3-14	98.89	98.89	98.89 ($\tau = 8$)	97.78 ($N = 10, \rho = 9 \times 10^{-6}$)
K3-23	98.89	97.78	98.89 ($\tau = 0$)	98.89 ($N = 10, \rho = 1 \times 10^{-6}$)
K3-24	97.78	95.56	97.78 ($\tau = 0$)	97.78 ($N = 8, \rho = 1 \times 10^{-6}$)
K3-34	92.22	86.67	95.56 ($\tau = 4$)	95.56 ($N = 6, \rho = 2 \times 10^{-6}$)
K6-12	73.33	53.33	65.00 ($\tau = 6$)	81.67 ($N = 6, \rho = 14 \times 10^{-6}$)
K6-13	71.67	60.00	68.33 ($\tau = 6$)	71.67 ($N = 8, \rho = 43 \times 10^{-6}$)
K6-14	91.67	80.00	95.00 ($\tau = 9$)	98.33 ($N = 10, \rho = 1 \times 10^{-6}$)
K6-23	78.33	61.67	78.33 ($\tau = 3$)	83.33 ($N = 6, \rho = 17 \times 10^{-6}$)
K6-24	81.67	70.00	81.67 ($\tau = 0$)	95.00 ($N = 10, \rho = 3 \times 10^{-6}$)
K6-34	96.67	73.33	96.67 ($\tau = 0$)	98.33 ($N = 10, \rho = 2 \times 10^{-6}$)
L1-12	90.00	80.00	93.33 ($\tau = 9$)	95.00 ($N = 6, \rho = 18 \times 10^{-6}$)
L1-13	90.00	81.67	91.67 ($\tau = 1$)	90.00 ($N = 8, \rho = 10 \times 10^{-6}$)
L1-14	96.67	75.00	98.33 ($\tau = 9$)	95.00 ($N = 6, \rho = 34 \times 10^{-6}$)
L1-23	93.33	83.33	93.33 ($\tau = 0$)	95.00 ($N = 10, \rho = 4 \times 10^{-6}$)
L1-24	93.33	86.67	95.00 ($\tau = 2$)	95.00 ($N = 6, \rho = 1 \times 10^{-6}$)
L1-34	75.00	70.00	78.33 ($\tau = 3$)	81.67 ($N = 8, \rho = 30 \times 10^{-6}$)
mean $\pm$ std	89.78 $\pm$ 0.10	80.03 $\pm$ 0.13	90.22 $\pm$ 0.11	92.72$\pm$0.08



*: $p < 0.05$ **: $p < 0.01$ ***: $p < 0.005$

Fig. 1: Weight maps of the optimized spatial filters for subset L1–12, associated with the three leading eigenvalues for each state (positioned in ascending order from left to right).

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